🎯 Lecture 25: Practice Problems — Technical Case Studies

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📌 Classification Practice Suite: Overview & Instructions

In machine learning applications, classification algorithms (such as Naive Bayes and k-Nearest Neighbors) serve as foundational operational engines. From clinical diagnostics to cybersecurity spam detection and industrial IoT telemetry, deploying production-grade systems requires far more than calling a generic `.fit()` method. Real-world engineering requires:

- Managing acute class imbalance with rigorous metric design (Confusion Matrix, Precision, Recall, Specificity, Macro and Weighted F1-Scores).
- Minimizing asymmetric real-world financial and life-safety penalty costs by calibrating optimal decision thresholds.
- Engineering mixed-type Bayesian generative classifiers handling discrete text tokens alongside continuous sensor features while eliminating zero-frequency pathology via Laplace smoothing.
- Navigating metric-space geometry in kNN, overcoming the Curse of Dimensionality, and eliminating insidious Data Leakage in Cross-Validation pipelines.

This suite features **3 comprehensive, textbook-grade industry case studies**:

1. **Case 1:** Medical Diagnostics Clinical Triaging & Cost-Sensitive Confusion Matrix Optimization
2. **Case 2:** Bayesian Spam Filtering & Gaussian vs Multinomial Naive Bayes with Laplace Smoothing
3. **Case 3:** Metric-Space Nearest Neighbor Classification (kNN), Curse of Dimensionality & Leakage-Free Stratified K-Fold Pipeline

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📌 Case Study 1: Medical Diagnostics Clinical Triaging & Cost-Sensitive Confusion Matrix Optimization

🏢 Real-World Context & Engineering Problem

A leading healthcare hospital network is deploying an automated AI screening diagnostic system designed to detect a rare, life-threatening cardiac arrhythmia condition from routine physiological telemetry (similar to features analyzed in our `heart.csv` dataset).

The hospital's screening cohort and diagnostic test parameters are formally defined as follows:

- **Total Patient Cohort:** The screening program processes $N = 10,000$ asymptomatic individuals per month:

$$N = 10,000$$

- **Disease Prevalence (Prior Probability):** The condition is rare across the general population, with a prevalence of 2:

$$P(D^+) = 0.02, \qquad P(D^-) = 1 - P(D^+) = 0.98$$

- **Diagnostic Model Sensitivity (Recall / True Positive Rate):** Clinical validation establishes a sensitivity of 95:

$$\text{Sensitivity} = \text{TPR} = P(T^+ \mid D^+) = \frac{TP}{P} = 0.95$$

- **Diagnostic Model Specificity (True Negative Rate):** Correct identification of healthy patients is certified at 90:

$$\text{Specificity} = \text{TNR} = P(T^- \mid D^-) = \frac{TN}{N_{\text{neg}}} = 0.90$$

- **False Positive Rate (FPR):** Probability of a false alarm on a healthy patient:

$$\text{FPR} = 1 - \text{Specificity} = 0.10$$

Hospital clinical leadership and data engineering face two paramount challenges:

1. **The Screening Paradox / Base Rate Fallacy:** When a patient tests positive (T^+), clinical intuition erroneously suggests a 95 probability of disease. The true posterior probability $P(D^+ \mid T^+)$ is substantially lower.
2. **Asymmetric Risk & Cost Penalties:** In clinical medicine, errors carry vastly unequal consequences:
   - **False Negative (FN) Cost:** A missed diagnosis allows a fatal cardiac event to occur untreated. Legal, remedial, and life-safety costs average C_{FN} =500$ per incident.
   - **False Positive (FP) Cost:** A false alarm results in a secondary non-invasive confirmation ultrasound costing C_{FP} =20$.
   - **Correct Classifications:** True Positives (C_{TP}) and True Negatives (C_{TN}) incur $$0$ penalty.

📋 What Is Being Asked in This Question? (Clear Objectives in Text Form)

You are tasked with providing the following **5 specific engineering deliverables**:

- **Bullet Point 1:** Construct the complete 2×2 Confusion Matrix (TP, FP, FN, TN) with exact integer counts for the $10,000$ patient cohort.
- **Bullet Point 2:** Apply Bayes' Theorem to derive and compute the Positive Predictive Value (PPV / Precision) $P(D^+ \mid T^+)$, providing a rigorous explanation of the Base Rate Fallacy.
- **Bullet Point 3:** Calculate Overall Accuracy, Macro F1-Score, and Weighted F1-Score. Mathematically prove why a "Dummy Baseline Classifier" predicting healthy for all patients attains 98 accuracy and why accuracy is dangerously misleading for medical triage.
- **Bullet Point 4:** Formulate the Cost-Sensitive Risk function, express expected clinical cost per patient, and analytically derive the Bayes optimal decision threshold τ^* that minimizes aggregate penalty loss.
- **Bullet Point 5:** Implement an executable Python script rendering a two-panel visualization: (a) Annotated Confusion Matrix heatmap, and (b) Expected Cost versus Decision Threshold curve highlighting the optimal operating cutoff.

🔍 Detailed Explanation of Each Bullet Point (Formulas & Theoretical Rationale)

1. Explanation of Bullet Point 1: Cohort Confusion Matrix Construction

- **Objective:** Quantify the exact partition of all $10,000$ screening subjects into the four fundamental outcomes of diagnostic classification.
- **Significance:** The confusion matrix represents the foundational truth table from which all classification metrics are derived.
- **Methodology:** Determine total diseased (P) and healthy (N_{neg}) sub-populations from prior prevalence:

$$P = N \cdot P(D^+), \qquad N_{\text{neg}} = N \cdot (1 - P(D^+))$$

Compute exact integer cell counts using sensitivity and specificity:

$$TP = P \cdot \text{Sensitivity}, \qquad FN = P - TP$$

$$TN = N_{\text{neg}} \cdot \text{Specificity}, \qquad FP = N_{\text{neg}} - TN$$

2. Explanation of Bullet Point 2: Precision & The Base Rate Fallacy

- **Objective:** Compute the true conditional probability of illness given a positive diagnostic report.

- **Significance:** Understanding why highly sensitive tests produce low precision under severe class imbalance is essential for clinical communication and engineering alert systems.
- **Methodology:** Expand Bayes' Theorem across the partitioned evidence:

$$P(D^+ \mid T^+) = \frac{P(T^+ \mid D^+)P(D^+)}{P(T^+)} = \frac{P(T^+ \mid D^+)P(D^+)}{P(T^+ \mid D^+)P(D^+) + P(T^+ \mid D^-)P(D^-)}$$

Cross-verify against the direct empirical ratio from the confusion matrix:

$$\text{Precision} = \frac{TP}{TP + FP}$$

3. Explanation of Bullet Point 3: Accuracy Paradox & F1-Score Breakdown

- **Objective:** Contrast naive aggregate accuracy against class-balanced harmonic metrics.
- **Significance:** Demonstrates the vulnerability of unweighted accuracy under extreme class distribution skew.
- **Methodology:** Compute aggregate accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Compute the disease class harmonic F1-score:

$$F1_{\text{disease}} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Evaluate Macro F1 (unweighted class average) and Weighted F1 (support-weighted average).

4. Explanation of Bullet Point 4: Cost-Sensitive Risk & Optimal Threshold τ^*

- **Objective:** Calibrate the probability decision boundary to minimize expected asymmetric loss.
- **Significance:** Machine learning models defaulting to $\tau = 0.50$ fail in asymmetric risk environments where missing a positive case is 25 times more costly than a false alarm.
- **Methodology:** Formulate total expected clinical risk:

$$\mathbb{E}[\text{Cost}] = C_{FN} \cdot FN + C_{FP} \cdot FP$$

By setting the expected marginal cost of positive classification equal to negative classification given posterior probability $p = P(D^+ \mid x)$, derive the Bayes optimal operating threshold:

$$\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

5. Explanation of Bullet Point 5: Python Simulation & Heatmap Visualizations

- **Objective:** Build an end-to-end Python pipeline illustrating triage metrics and cost curves.
- **Significance:** Visual validation bridges theoretical derivation and clinical operations.
- **Methodology:** Use `matplotlib` to render a labeled confusion matrix heatmap and a continuous cost trace sweeping $\tau \in [0.001, 0.50]$.

📌 Case Study 2: Bayesian Spam Filtering & Gaussian vs Multinomial Naive Bayes with Laplace Smoothing

🏢 Real-World Context & Engineering Problem

An enterprise email gateway processes high-throughput email traffic to block phishing scams and malware distribution while ensuring zero legitimate business communications (ham) are dropped.

The security gateway captures two complementary feature spaces:

1. **Discrete Word Token Occurrences:** Specific lexical trigger terms modeled via Multinomial Naive Bayes.
2. **Continuous Metadata Metric:** An aggregate urgency score $x_{\text{urgency}} \in \mathbb{R}$, combining uppercase character density, punctuation frequency, and server dispatch rates, modeled via Gaussian Naive Bayes.

Training data comprises $N = 1,000$ verified emails:

- **Total Corpus Size:**

$$N = 1,000$$

- **Class Priors:**

$$N_{\text{spam}} = 300, \qquad N_{\text{ham}} = 700$$

$$P(\text{Spam}) = \frac{300}{1000} = 0.30, \qquad P(\text{Ham}) = \frac{700}{1000} = 0.70$$

- **Lexicon Characteristics:** Vocabulary size $|V| = 500$ distinct tokens.
  - Total token occurrences in spam corpus: $T_{\text{spam}} = 3,000$.
  - Total token occurrences in ham corpus: $T_{\text{ham}} = 7,000$.
- **Word Frequency Table for Critical Terms:**

Keyword (w_j)	Spam Corpus Count ($\text{count}(w_j, S)$)	Ham Corpus Count ($\text{count}(w_j, H)$)
$w_1 = \text{"lottery"}$	120	2
$w_2 = \text{"winner"}$	90	5
$w_3 = \text{"meeting"}$	15	420
$w_4 = \text{"cryptocurrency"}$	10	0

- **Continuous Urgency Feature Parameters (x_{urgency}):**
  - Spam class distribution follows Gaussian $\mathcal{N}(\mu_S, \sigma_S^2)$:

$$\mu_S = 8.5, \qquad \sigma_S = 1.5 \implies \sigma_S^2 = 2.25$$

- Ham class distribution follows Gaussian $\mathcal{N}(\mu_H, \sigma_H^2)$:

$$\mu_H = 2.0, \qquad \sigma_H = 1.0 \implies \sigma_H^2 = 1.00$$

An incoming suspicious test email E_{test} arrives with keywords:

$$\text{"lottery"}, \text{"winner"}, \text{"cryptocurrency"}$$

and urgency metric:

$$x_{\text{urgency}} = 6.0$$

📋 What Is Being Asked in This Question? (Clear Objectives in Text Form)

You are tasked with providing the following **5 specific engineering deliverables**:

- **Bullet Point 1:** Formulate the Conditional Independence Assumption of Naive Bayes and explain why joint likelihood simplifies to a product of marginal likelihoods.
- **Bullet Point 2:** Define the Zero-Frequency Problem, prove why an unregularized MLE calculation for $w_4 = \text{"cryptocurrency"}$ forces $P(\text{Ham} \mid E) = 0$, derive Laplace Add-1 Smoothing, and compute smoothed conditional probabilities for all four terms across both classes.
- **Bullet Point 3:** Formulate the Log-Likelihood objective and prove how transformation to log-space prevents floating-point underflow under 64-bit IEEE-754 arithmetic.
- **Bullet Point 4:** State the Gaussian Naive Bayes PDF, compute class-conditional probability densities for urgency score $x = 6.0$, and evaluate the urgency likelihood ratio.
- **Bullet Point 5:** Synthesize discrete lexical likelihoods, continuous Gaussian density, and prior probabilities to evaluate posterior probabilities $P(\text{Spam} \mid E_{\text{test}})$ and $P(\text{Ham} \mid E_{\text{test}})$, issuing the final gateway verdict.

🔍 Detailed Explanation of Each Bullet Point (Formulas & Theoretical Rationale)

1. Explanation of Bullet Point 1: Conditional Independence Assumption

- **Objective:** Deconstruct the foundational statistical assumption powering Naive Bayes classification.
- **Significance:** Explains how linear parameter scaling $\mathcal{O}(d \cdot K)$ replaces exponential parameter explosion $\mathcal{O}(2^d \cdot K)$.
- **Methodology:** Express the decomposition of the joint class-conditional likelihood:

$$P(\mathbf{x} \mid C_k) = P(x_1, x_2, \ldots, x_d \mid C_k) = \prod_{j=1}^{d} P(x_j \mid C_k)$$

2. Explanation of Bullet Point 2: Zero-Frequency Problem & Laplace Smoothing ($\alpha = 1$)

- **Objective:** Prevent catastrophic probability collapse when encountering novel or unobserved vocabulary in test instances.
- **Significance:** Without smoothing, a single unseen token annihilates evidence from dozens of highly informative tokens.
- **Methodology:** State the Laplace smoothed estimator:

$$\hat{P}(w_j \mid C_k) = \frac{\text{count}(w_j, C_k) + \alpha}{\sum_{w \in V} \text{count}(w, C_k) + \alpha |V|}$$

Apply $\alpha = 1$, $|V| = 500$, $T_{\text{spam}} = 3000$, and $T_{\text{ham}} = 7000$ to evaluate all word conditional probabilities.

3. Explanation of Bullet Point 3: Log-Likelihood Formulation & Underflow Prevention

- **Objective:** Eliminate IEEE-754 floating point arithmetic underflow in high-dimensional feature spaces.
- **Significance:** Products of hundreds of small fractional probabilities rapidly drop below double-precision limit (10^{-308}).
- **Methodology:** Apply monotonic logarithmic transformation:

$$\log P(C_k \mid \mathbf{x}) \propto \log P(C_k) + \sum_{j=1}^{d} \log P(x_j \mid C_k)$$

4. Explanation of Bullet Point 4: Continuous Features via Gaussian Naive Bayes PDF

- **Objective:** Incorporate continuous real-valued metrics alongside discrete word frequency counts.
- **Significance:** Allows hybrid systems to leverage continuous telemetry without arbitrary binning or quantization.
- **Methodology:** Evaluate the univariate Gaussian probability density function:

$$f(x \mid \mu_k, \sigma_k^2) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right)$$

Compute densities for Ham ($\mu = 2.0, \sigma = 1.0$) and Spam ($\mu = 8.5, \sigma = 1.5$) at $x = 6.0$.

5. Explanation of Bullet Point 5: Joint Posterior Inference & Decision Rule

- **Objective:** Synthesize multi-modal evidence into an actionable classification verdict.
- **Significance:** Demonstrates full-system Bayesian reasoning from priors and disparate likelihoods to calibrated posterior odds.
- **Methodology:** Compute unnormalized class scores:

$$\text{Score}(\text{Spam}) = P(\text{Spam}) \cdot \left(\prod_{w \in E_{\text{test}}} \hat{P}(w \mid \text{Spam}) \right) \cdot f(6.0 \mid \text{Spam})$$

$$\text{Score}(\text{Ham}) = P(\text{Ham}) \cdot \left(\prod_{w \in E_{\text{test}}} \hat{P}(w \mid \text{Ham}) \right) \cdot f(6.0 \mid \text{Ham})$$

Normalize to obtain posterior percentages and apply the maximum a posteriori (MAP) decision rule.

📌 Case Study 3: Metric-Space Nearest Neighbor Classification (kNN), Curse of Dimensionality & Leakage-Free Stratified K-Fold Pipeline

🏗️ Real-World Context & Engineering Problem

An industrial automation manufacturing facility utilizes edge telemetry IoT sensors mounted on high-speed rotational turbines. The predictive maintenance classification pipeline monitors real-time telemetry to classify turbine operational health as either Normal ($y = 0$) or Faulty ($y = 1$).

Sensors capture two physical continuous telemetry dimensions:

- **Feature 1:** Bearing Operating Temperature x_1 ($^\circ\text{C}$), spanning range $[20, 100]$ ($\Delta x_1 \approx 80$).
- **Feature 2:** Micro-vibration Amplitude x_2 (mm), spanning range $[0.001, 0.050]$ ($\Delta x_2 \approx 0.049$).

Six historical calibration baseline samples are stored in the edge memory index:

$$P_1 = [25.0, 0.005]^T, \quad y_1 = 0 \quad (\text{Normal}) \; P_2 \quad = [30.0, 0.008]^T, \quad y_2 = 0 \quad (\text{Normal}) \; P_3 = [35.0, 0.012]^T, \quad y_3 = 0 \quad (\text{No}$$

A live telemetry observation query Q is received from turbine unit #7:

$$Q = [40.0, 0.042]^T$$

Crucially, query Q registers moderate temperature (40°C, close to normal operating baselines) but hazardous micro-vibration (0.042 mm, characteristic of impending bearing seizure).

Engineering leadership mandates addressing two algorithmic risks:

1. **Metric Space Distortion from Unstandardized Scales:** Raw distance calculation causes the high-magnitude temperature scale to completely overshadow vibration amplitude.
2. **Data Leakage in Cross-Validation:** Performing global preprocessing transformations before dataset splitting leaks test set distributions into training splits, yielding over-optimistic evaluation metrics that fail in deployment.

📋 What Is Being Asked in This Question? (Clear Objectives in Text Form)

You are tasked with providing the following **5 specific engineering deliverables**:

- **Bullet Point 1:** Calculate raw unscaled Euclidean distances from query Q to all reference points $P_1 \dots P_6$. Prove mathematically that temperature differences contribute > 99.99 of the total distance metric, completely blinding the classifier to vibration.
- **Bullet Point 2:** Calculate reference sample mean vector $\boldsymbol{\mu}$ and standard deviation vector $\boldsymbol{\sigma}$. Perform Z-score standardization on all points and compute standardized Euclidean distances. Prove that the top 3 nearest neighbors completely invert between unscaled and scaled spaces.
- **Bullet Point 3:** Formulate Uniform Majority Voting and Inverse-Distance Weighted Voting schemes, computing the predicted class label for query Q under both rules for $k = 3$ and $k = 5$.
- **Bullet Point 4:** Formulate the Curse of Dimensionality and the Metric Collapse Theorem, proving mathematically why relative distance contrast collapses to zero as dimensionality $d \to \infty$ ($\lim_{d \to \infty} \frac{D_{\max} - D_{\min}}{D_{\min}} \to 0$).
- **Bullet Point 5:** Explain the mechanism of Data Leakage during feature scaling and construct a production-ready Scikit-Learn `Pipeline` pairing `StandardScaler` and `KNeighborsClassifier` inside `StratifiedKFold(n_splits=5)` to guarantee zero leakage during cross-validation.

🔍 Detailed Explanation of Each Bullet Point (Formulas & Theoretical Rationale)

1. Explanation of Bullet Point 1: Unscaled Euclidean Distances & Scale Distortion

- **Objective:** Quantify how differences in feature measurement units distort geometric proximity.

- **Significance:** Metric-space classifiers measure similarity via geometric distance; arbitrary scale disparities destroy physical meaning.
- **Methodology:** Compute standard Euclidean distance:

$$d(Q, P_i) = \sqrt{(x_{Q,1} - x_{i,1})^2 + (x_{Q,2} - x_{i,2})^2}$$

Compute the proportional variance contribution of dimension 1 versus dimension 2.

2. Explanation of Bullet Point 2: Z-Score Standardization & Neighbor Inversion

- **Objective:** Equalize feature dimensional contributions to geometric distance.
- **Significance:** Standardization ensures each feature contributes proportionally to unit variance.
- **Methodology:** Evaluate sample moments across the training set:

$$\mu_j = \frac{1}{n}\sum_{i=1}^{n} x_{i,j}, \qquad \sigma_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_{i,j} - \mu_j)^2}$$

Standardize reference points and query point:

$$z_{i,j} = \frac{x_{i,j} - \mu_j}{\sigma_j}, \qquad z_{Q,j} = \frac{x_{Q,j} - \mu_j}{\sigma_j}$$

Recompute standardized Euclidean distances:

$$d_{\text{scaled}}(Q, P_i) = \sqrt{\sum_{j=1}^{2}(z_{Q,j} - z_{i,j})^2}$$

3. Explanation of Bullet Point 3: Uniform vs Distance-Weighted Voting

- **Objective:** Compare decision aggregation rules across nearest neighborhood sets.
- **Significance:** Distance weighting prevents distant outliers from outvoting immediate proximate neighbors.
- **Methodology:** Compute uniform majority prediction:

$$\hat{y} = \text{mode}(y_i : P_i \in \mathcal{N}_k(Q))$$

Compute inverse distance weighted prediction:

$$w_i = \frac{1}{d_{\text{scaled}}(Q, P_i) + \epsilon}, \qquad \hat{y} = \arg\max_{c \in 0,1} \sum_{i \in \mathcal{N}_k(Q), y_i = c} w_i$$

4. Explanation of Bullet Point 4: Curse of Dimensionality & Metric Collapse

- **Objective:** Mathematically analyze nearest neighbor degradation in high-dimensional feature spaces.
- **Significance:** Guides feature selection and dimensionality reduction (PCA) prior to distance-based modeling.
- **Methodology:** Analyze distance variance as $d \to \infty$ for i.i.d. random variables:

$$\lim_{d \to \infty} \frac{D_{\max} - D_{\min}}{D_{\min}} \to 0$$

Explain why the concept of "nearest" neighbor becomes mathematically indistinguishable from any arbitrary point.

5. Explanation of Bullet Point 5: Data Leakage Prevention via Scikit-Learn Pipelines

- **Objective:** Implement rigorous cross-validation isolation for data-dependent transformations.
- **Significance:** Prevents optimistic cross-validation score inflation caused by test set leakage.
- **Methodology:** Encapsulate preprocessing and estimation within `sklearn.pipeline.Pipeline`:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import StratifiedKFold, cross_val_score

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('knn', KNeighborsClassifier(n_neighbors=5, weights='distance'))
])

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(pipeline, X, y, cv=cv, scoring='f1_macro')
```